SCIENCE (Code 086) — Marking Scheme Excerpt

CLASS X (2025-26) — Test subset: Q10, Q13, Q28

(Extracted from the official CBSE Class X Science Marking Scheme 2025-26 — this is your reference_answer + marking_synoptic)

Q10. **2 marks**

It is completely wrong to say that plants do not produce any excretory products. However, plants use completely different strategies for excretion than those of the animals. They get rid of these wastes in different manner (any two):

- Oxygen, a photosynthetic waste, is removed through stomata.
- Excess water is removed by transpiration through stomata.
- Other metabolic wastes are either stored in dead cells, resins and gums, or removed through falling of old leaves.
- Many waste products are stored in cellular vacuoles.

Marking synoptic: 1 mark for correcting the statement, 1 mark for any two valid mechanisms (each mechanism named).

Q13. **3 marks**

All information from our environment is detected by the specialised tips of some nerve cells. The information acquired at the end of the dendritic tip of a nerve cell sets off a chemical reaction that creates an electrical impulse. This impulse travels from the dendrite to the cell body, and then along the axon to its end. At the end of the axon, the electrical impulse sets off the release of some chemicals. These chemicals cross the gap, or synapse, and start a similar electrical impulse in a dendrite of the next neuron. This is how nervous impulses travel in the body. **(Diagram of neuron — dendrite, cell body, axon, nerve ending — required.)**

Marking synoptic: 1 mark for dendrite → electrical impulse; 1 mark for impulse travelling via axon to synapse and chemical release; 1 mark for labelled diagram of the neuron.

Q28. **4 marks**

A. (b) < 40, because concentrated H_2SO_4 gives more H+ ions than dilute acid.

B. 3 mL of H_2SO_4 will be 60 drops, which will neutralise 6 mL of NaOH.

C. $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$ — (a) neutralisation and double displacement reaction. Base NaOH is getting neutralised and forming salt + water. It is double displacement as Na+ ions are being replaced by H+ and OH- by $SO4^{2-}$. It is not a precipitation reaction because Na_2SO_4 is soluble in water.

Marking synoptic: 1 mark for part A with correct reasoning; 1 mark for part B numeric answer; 1 mark for balanced equation; 1 mark for correctly identifying + explaining the reaction type.